Ibn Tofail University

Algebra IV

Problem Set I

Exercise 1:

Let $f \in \text{End}(\mathbb{C}^3)$ with matrix relative to the canonical basis:

$$A = \begin{pmatrix} 2 & 0 & 4 \\ 3 & -4 & 12 \\ 1 & -2 & 5 \end{pmatrix}$$

1. Find the eigenvalues of f.
2. Show there exists a basis B of eigenvectors of f and give the matrix of f in B.
3. Give the change of basis matrix P from the canonical basis B_0 to B, then compute $P^{-1}AP$.

Correction

1. Finding the eigenvalues

$$\det(A - \lambda I) = \lambda(2 - \lambda)(\lambda - 1)$$

2. Finding eigenvectors and basis B

For $\lambda = 0$: Solve $A\mathbf{v} = 0$

$$\Rightarrow x = -2z, \; y = \frac{3}{2}z$$

Take $z = 2$: $v_1 = (-4, 3, 2)$

For $\lambda = 1$: Solve $(A - I)\mathbf{v} = 0$

$$\Rightarrow x = -4z, \; y = 0$$

Take $z = 1$: $v_2 = (-4, 0, 1)$

For $\lambda = 2$: Solve $(A - 2I)\mathbf{v} = 0$

$$\Rightarrow z = 0, \; x = 2y$$

Take $y = 1$: $v_3 = (2, 1, 0)$

Basis $B = \{v_1, v_2, v_3\} = \{(-4, 3, 2), (-4, 0, 1), (2, 1, 0)\}$

Matrix of f in basis B:

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

3. Change of basis matrix and similarity transformation

   Change of basis matrix:
$$P = \begin{pmatrix} -4 & -4 & 2 \\ 3 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$
   We verify that:
$$P^{-1}AP = D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$
   This confirms that A is diagonalizable and B is a basis of eigenvectors.

Exercise 2:

Let
$$A = \begin{pmatrix} 2 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$
Find the characteristic polynomial of A.

Correction

The characteristic polynomial is given by:
$$P(\lambda) = \det(A - \lambda I)$$
where I is the 3×3 identity matrix.
We compute, Using the cofactor expansion along the first row:
$$\begin{aligned} \det(A - \lambda I) &= (2-\lambda)\left[(-\lambda)(1-\lambda) - (1)(1)\right] \\ &\quad - (-1)\left[(2)(1-\lambda) - (1)(1)\right] \\ &\quad + (1)\left[(2)(1) - (-\lambda)(1)\right] \\ \iff P(\lambda) &= (2-\lambda)(\lambda^2 - \lambda - 1) + (1 - 2\lambda) + (2 + \lambda) \\ &= -\lambda^3 + 3\lambda^2 - 2\lambda + 1 \end{aligned}$$

Exercise 3:

Let $A \in M_K$ with characteristic polynomial P such that $P(0) \neq 0$.

- Show A is invertible and give the expression for the characteristic polynomial of A.
- Show for all $B \in M_K$, AB and BA have the same characteristic polynomial.

Correction

1. Since $P(\lambda) = \det(A - \lambda I)$, we have $P(0) = \det(A)$. The hypothesis $P(0) \neq 0$ implies $\det(A) \neq 0$, so A is invertible.

   Let $Q(\lambda)$ be the characteristic polynomial of A^{-1}. Then:

   $$Q(\lambda) = \det(A^{-1} - \lambda I) = \det(A^{-1}(I - \lambda A)) = \det(A^{-1})\det(I - \lambda A).$$

   Since $\det(A^{-1}) = 1/\det(A) = 1/P(0)$ and

   $$\det(I - \lambda A) = \det(-\lambda(A - \lambda^{-1} I)) = (-\lambda)^n \det(A - \lambda^{-1} I) = (-\lambda)^n P(\lambda^{-1}),$$

   we obtain

   $$Q(\lambda) = \frac{1}{P(0)} \cdot (-\lambda)^n P(\lambda^{-1}) = (-1)^n \frac{\lambda^n}{P(0)} P\left(\frac{1}{\lambda}\right).$$

2. Since A is invertible (from the first part), for any $B \in M_n(K)$ we have

   $$BA = A^{-1}(AB)A,$$

   which shows that AB and BA are similar matrices. Similar matrices have the same characteristic polynomial, so AB and BA share the same characteristic polynomial.

Exercise 4:

Given

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix}$$

1. Determine the characteristic polynomial of A.
2. Determine eigenvalues and give a basis for each eigenspace of A.
3. Is A triangulable? Diagonalizable?

Correction

1. The characteristic polynomial of A is $P(\lambda) = \det(A - \lambda I)$.

   $$\begin{aligned} P(\lambda) &= (3-\lambda)\det\begin{pmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{pmatrix} - 1 \cdot \det\begin{pmatrix} 2 & 2 \\ 1 & 3-\lambda \end{pmatrix} + 1 \cdot \det\begin{pmatrix} 2 & 4-\lambda \\ 1 & 1 \end{pmatrix} \\ &= -(\lambda - 2)^2(\lambda - 6). \end{aligned}$$

2. The eigenvalues are the roots of $P(\lambda)$: $\lambda_1 = 2$ (algebraic multiplicity 2) and $\lambda_2 = 6$ (algebraic multiplicity 1). **Eigenspace for $\lambda = 2$:** Solve $(A - 2I)v = 0$:

   $$A - 2I = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{row reduction}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The equation is $x + y + z = 0$, so $x = -y - z$. A basis is:

$$\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Eigenspace for $\lambda = 6$: Solve $(A - 6I)v = 0$:

$$A - 6I = \begin{pmatrix} -3 & 1 & 1 \\ 2 & -2 & 2 \\ 1 & 1 & -3 \end{pmatrix} \xrightarrow{\text{row reduction}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The equations are $x = z$ and $y = 2z$. A basis is:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

The geometric multiplicity is 1.

3. Since all eigenvalues are real (assuming $K = \mathbb{R}$ or $\mathbb{C}$) and the characteristic polynomial splits into linear factors, A is triangulable over K.

Exercise 5:

Let $n \in \mathbb{N}$, K a commutative field, and $A \in M$.

- If $\operatorname{rank}(A) = 1$, show A is diagonalizable if and only if $\operatorname{tr}(A) = 0$.

Correction

Let $n \in \mathbb{N}$, K a commutative field, and $A \in M_n(K)$ with $\operatorname{rank}(A) = 1$.
Since $\operatorname{rank}(A) = 1$, the image of A is a one-dimensional subspace of K^n. Therefore, there exist non-zero vectors $u, v \in K^n$ such that

$$A = uv^\top,$$

where $v^\top$ denotes the transpose of v (so $A_{ij} = u_i v_j$).
The characteristic polynomial of a rank-1 matrix has a well-known form. Indeed, for any $\lambda \in K$,

$$\det(A - \lambda I) = (-\lambda)^{n-1}\big(\operatorname{tr}(A) - \lambda\big).$$

This can be seen by noting that 0 is an eigenvalue of A with geometric multiplicity at least $n-1$ (since $\dim\ker A = n - \operatorname{rank}(A) = n-1$), and the remaining eigenvalue must be $\operatorname{tr}(A)$, because the trace equals the sum of all eigenvalues (counted with algebraic multiplicities).
Thus, the eigenvalues of A are:

$$\lambda_1 = \operatorname{tr}(A) \quad \text{(algebraic multiplicity 1)}, \qquad \lambda_2 = 0 \quad \text{(algebraic multiplicity } n-1\text{)}.$$

Now, A is diagonalizable if and only if, for each eigenvalue, its geometric multiplicity equals its algebraic multiplicity.

- The eigenspace for $\lambda = 0$ is $\ker A$, which has dimension $n - \operatorname{rank}(A) = n - 1$. Hence, the geometric multiplicity of 0 is $n-1$, which already matches its algebraic multiplicity.
- For $\lambda = \operatorname{tr}(A)$, the geometric multiplicity is 1 if $\operatorname{tr}(A) \neq 0$ (since the image of A is one-dimensional and consists of eigenvectors with eigenvalue $\operatorname{tr}(A)$), and 0 if $\operatorname{tr}(A) = 0$ (because then $\operatorname{tr}(A) = 0$ is not a distinct eigenvalue).
We now consider two cases:
Case 1: $\operatorname{tr}(A) \neq 0$**.** Then the eigenvalues are $\operatorname{tr}(A)$ (multiplicity 1) and 0 (multiplicity $n-1$). The geometric multiplicity of $\operatorname{tr}(A)$ is 1 (since $\operatorname{Im}(A) \subseteq E_{\operatorname{tr}(A)}$ and $\dim \operatorname{Im}(A) = 1$), so all geometric multiplicities match algebraic multiplicities. Hence, A is diagonalizable.
Case 2: $\operatorname{tr}(A) = 0$**.** Then the only eigenvalue is 0, with algebraic multiplicity n. However, the geometric multiplicity of 0 is $\dim \ker A = n - 1 < n$. Therefore, A is **not** diagonalizable.
But wait—this contradicts the statement we are asked to prove. The issue is in the interpretation: the correct statement is actually:

$$\text{If } \operatorname{rank}(A) = 1, \text{ then } A \text{ is diagonalizable} \iff \operatorname{tr}(A) \neq 0.$$

However, the problem claims the equivalence with $\operatorname{tr}(A) = 0$. This suggests either a typo in the problem or a different convention.
Let us double-check with an example:
- Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then $\operatorname{rank}(A) = 1$, $\operatorname{tr}(A) = 1 \neq 0$, and A is diagonal (hence diagonalizable).
- Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $\operatorname{rank}(A) = 1$, $\operatorname{tr}(A) = 0$, but A is a non-zero nilpotent matrix, so it is not diagonalizable.
Thus, the correct equivalence is:

$$A \text{ is diagonalizable} \iff \operatorname{tr}(A) \neq 0.$$

Therefore, the statement in the problem appears to contain an error. Assuming the intended statement is the correct one, we conclude:

$$\boxed{A \text{ is diagonalizable if and only if } \operatorname{tr}(A) \neq 0.}$$

If, however, the problem insists on the given formulation, then it is false as stated.

Exercise 6:

Show matrix $A \in M_K$ with entries $a_{ij} = 1$ for all i, j is diagonalizable.
Determine eigenvalues and eigenspaces of the linear map defined by

$$f(X) = X \cdot (PX_1X - 3P - XP)$$

Correction

We address the two parts of the problem separately.

Part 1: Diagonalizability of the all-ones matrix.
Let $A \in M_n(K)$ be the matrix defined by $a_{ij} = 1$ for all i, j. This is the rank-1 matrix

$$A = \mathbf{1}\,\mathbf{1}^\top,$$

where $\mathbf{1} = (1, 1, \dots, 1)^\top \in K^n$.
Since $\operatorname{rank}(A) = 1$ (all rows are identical and non-zero), we know from standard results that:

1. 0 is an eigenvalue with geometric multiplicity $n-1$ (because $\dim \ker A = n - \operatorname{rank}(A) = n - 1$).
2. The trace of A is $\operatorname{tr}(A) = \sum_{i=1}^n a_{ii} = n$.

For a rank-1 matrix, the non-zero eigenvalue (if any) equals the trace. Hence the eigenvalues are:

$$\lambda_1 = n \quad \text{(algebraic multiplicity 1)}, \qquad \lambda_2 = 0 \quad \text{(algebraic multiplicity } n-1\text{)}.$$

Now, the eigenspace for $\lambda = n$ is $\operatorname{Im}(A) = \operatorname{span}\{\mathbf{1}\}$, which is 1-dimensional. The eigenspace for $\lambda = 0$ is $\ker A = \{x \in K^n \mid x_1 + x_2 + \cdots + x_n = 0\}$, which has dimension $n-1$.
Thus, the sum of the geometric multiplicities is $1+(n-1) = n$, so A is diagonalizable (provided that $n \neq 0$ in K; i.e., the characteristic of K does not divide n). In particular, over $\mathbb{R}$ or $\mathbb{C}$, A is always diagonalizable.

Part 2: Eigenvalues and eigenspaces of $f(X) = X \cdot (PX_1X - 3P - XP)$.
The expression for the linear map f is ambiguous and likely contains typographical errors. The notation $X \cdot (PX_1X - 3P - XP)$ is not standard for a linear map on a space of matrices or vectors:

1. It is unclear what X_1 denotes (a fixed matrix? a coordinate?).
2. The term PX_1X suggests a product of three objects, but without knowing the dimensions or nature of P, X_1, and X, the expression is undefined.
3. Moreover, the map as written is generally **not linear** in X, due to the presence of quadratic terms like XPX or PX_1X.

Since the problem states that f is a linear map, the given formula must be incorrect or miswritten. A common linear map involving a fixed matrix P is, for example:

$$f(X) = PX - XP \quad \text{(the commutator map)},$$

or

$$f(X) = PX + XP,$$

both of which are linear in X.
Without a clear and mathematically consistent definition of f, it is impossible to determine its eigenvalues and eigenspaces.

Conclusion:

1. The all-ones matrix is diagonalizable (over fields where $n \neq 0$), with eigenvalues n (multiplicity 1) and 0 (multiplicity $n-1$).
2. The second part cannot be solved as stated due to ambiguous or erroneous notation.

Exercise 7:

For $n \in \mathbb{N}$ and two $n \times n$ matrices A, B, show every eigenvalue of AB is an eigenvalue of BA.

Correction

Let $n \in \mathbb{N}$ and let $A, B \in M_n(K)$. We show that every eigenvalue of AB is also an eigenvalue of BA.
Let $\lambda \in K$ be an eigenvalue of AB. Then there exists a non-zero vector $v \in K^n$ such that

$$ABv = \lambda v.$$

We consider two cases.

Case 1: $\lambda \neq 0$.
Since $\lambda \neq 0$ and $v \neq 0$, we have $Bv \neq 0$ (otherwise $ABv = A0 = 0 = \lambda v$ would imply $\lambda v = 0$, contradicting $\lambda \neq 0$ and $v \neq 0$).
Now apply B to both sides of $ABv = \lambda v$:

$$B(ABv) = B(\lambda v) \quad \Longrightarrow \quad (BA)(Bv) = \lambda(Bv).$$

Since $Bv \neq 0$, this shows that Bv is an eigenvector of BA with eigenvalue λ. Hence λ is an eigenvalue of BA.

Case 2: $\lambda = 0$.
We must show that if 0 is an eigenvalue of AB, then it is also an eigenvalue of BA. This is equivalent to showing that if AB is singular, then BA is singular.
Recall that for square matrices, $\det(AB) = \det(A)\det(B) = \det(BA)$. Therefore,

$$\det(AB) = 0 \quad \Longleftrightarrow \quad \det(BA) = 0.$$

Thus, AB is not invertible if and only if BA is not invertible, which means 0 is an eigenvalue of AB if and only if 0 is an eigenvalue of BA.

In both cases, every eigenvalue λ of AB is also an eigenvalue of BA. (In fact, the same argument with A and B swapped shows the converse, so AB and BA have exactly the same eigenvalues, including algebraic multiplicities when K is algebraically closed.)
Therefore, every eigenvalue of AB is an eigenvalue of BA.

Exercise 8:

For what values of real parameters a, b, c, d, e, f are the following matrices diagonalizable?

$$1. \quad A = \begin{pmatrix} a & b & c \\ 0 & 2 & d \\ e & 0 & 2 \end{pmatrix} \qquad 2. \quad B = \begin{pmatrix} a & b & c \\ 0 & 1 & d \\ e & 0 & 2 \end{pmatrix}$$

Correction

The matrix

$$A = \begin{pmatrix} a & b & c \\ 0 & 2 & d \\ e & 0 & 2 \end{pmatrix}$$

is diagonalizable over $\mathbb{R}$ **if and only if**:

$$e = 0 \quad \text{and} \quad \begin{cases} d = 0 & \text{if } a \neq 2 \quad (\text{with } b, c \text{ arbitrary}), \\ b = c = d = 0 & \text{if } a = 2. \end{cases}$$

In short: - e must be zero (no coupling from row 3 to column 1), - and the lower 2×2 block $\begin{pmatrix} 2 & d \\ 0 & 2 \end{pmatrix}$ must be diagonalizable, which forces $d = 0$, - unless $a = 2$, in which case the whole matrix must be diagonal (so $b = c = d = 0$).

Exercise 9:

Given

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 2 & 0 & -1 \end{pmatrix}$$

Use Cayley-Hamilton theorem to show A is invertible and compute its inverse.

Correction

Step 1: Characteristic polynomial. Compute $P(\lambda) = \det(A - \lambda I)$: Expand along the third column (only one non-zero entry):

$$P(\lambda) = (-1 - \lambda) \cdot \det \begin{pmatrix} 1 - \lambda & 1 \\ -1 & -\lambda \end{pmatrix} = (-1 - \lambda)\big[(1 - \lambda)(-\lambda) - (-1)(1)\big].$$

So the characteristic polynomial is

$$P(\lambda) = -\lambda^3 + 0\lambda^2 + 0\lambda + 1 = -\lambda^3 + 1,$$

or equivalently (multiplying by -1, which doesn't affect Cayley–Hamilton):

$$\chi_A(\lambda) = \lambda^3 - 1.$$

Step 2: Apply Cayley–Hamilton. The theorem states $\chi_A(A) = 0$, so:

$$A^3 - I = 0 \quad \Longrightarrow \quad A^3 = I.$$

Step 3: Invertibility and inverse. From $A^3 = I$, multiply both sides by A^{-1} (if it exists) to get $A^2 = A^{-1}$. But more directly:

$$A \cdot A^2 = A^3 = I,$$

so A is invertible and

$$A^{-1} = A^2.$$

Step 4: Compute A^2.

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 2 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 2 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix}.$$

Answer: A is invertible and

$$A^{-1} = A^2 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix}.$$

Exercise 10:

Let E be a vector space over K of dimension n, and u an endomorphism of E.
Show that for all $p \in \mathbb{N}$, $u^p \in Vect(I_d, u, \ldots, u^{m-1})$.
Application: Let A be the matrix

$$\begin{pmatrix} 2 & 0 & 4 \\ 3 & -4 & 12 \\ 1 & -2 & 5 \end{pmatrix}$$

For $p \in \mathbb{N}$, write A^p as a linear combination of I, A, A^2.

Correction

Let E be a vector space over K with $\dim E = n$, and $u \in \mathrm{End}(E)$. Since $\dim \mathrm{End}(E) = n^2$, the $n^2 + 1$ endomorphisms $I, u, u^2, \ldots, u^{n^2}$ are linearly dependent. Hence there exists a non-zero polynomial P such that $P(u) = 0$. Let $m = \deg(\mu_u) \le n$ be the degree of the minimal polynomial of u. Then

$$u^m \in \mathrm{Vect}(I, u, \ldots, u^{m-1}),$$

and by induction, $u^p \in \mathrm{Vect}(I, u, \ldots, u^{m-1})$ for all $p \in \mathbb{N}$.

Application. Let

$$A = \begin{pmatrix} 2 & 0 & 4 \\ 3 & -4 & 12 \\ 1 & -2 & 5 \end{pmatrix}.$$

Compute the characteristic polynomial:

$$\chi_A(\lambda) = \det(A - \lambda I) = -\lambda(\lambda - 1)(\lambda - 2).$$

By the Cayley–Hamilton theorem, $\chi_A(A) = 0$, so

$$A^3 - 3A^2 + 2A = 0 \quad \Longrightarrow \quad A^3 = 3A^2 - 2A.$$

This recurrence allows us to reduce any higher power of A to a linear combination of I, A, and A^2. In fact, since the relation contains no I term, for $p \geq 1$ the powers lie in $\mathrm{Vect}(A, A^2)$, but we include I for $p = 0$.
Thus, for every $p \in \mathbb{N}$, there exist scalars $\alpha_p, \beta_p, \gamma_p \in K$ such that

$$A^p = \alpha_p I + \beta_p A + \gamma_p A^2,$$

with the recurrence (for $p \geq 3$):

$$A^p = 3A^{p-1} - 2A^{p-2}.$$